Midterm Project Report - NewsBot Intelligence System

From News Classification to Explainable News Intelligence

Course: ITAI 2373, Natural Language Processing

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GitHub Repository: https://github.com/iamhuongthn/ITAI2373-NewsBot-Midterm

1. Executive Summary

News articles contain more information than their assigned categories. A business article may discuss economic uncertainty, a technology article may introduce emerging companies, and a political article may contain multiple organizations, individuals, and relationships. Simply identifying the topic is useful, but it does not fully explain what an article communicates.

I developed **NewsBot Intelligence System**, an end-to-end Natural Language Processing (NLP) prototype that combines eight NLP modules to transform unstructured news articles into structured, interpretable information.

The system integrates text preprocessing, TF-IDF feature extraction, part-of-speech analysis, dependency parsing, sentiment and emotion analysis, multi-class classification, and named entity recognition.

Using the BBC News Classification Dataset, I prepared 1,440 unique articles across five categories: Business, Entertainment, Politics, Sport, and Technology.

Three machine learning configurations were evaluated: Multinomial Naive Bayes, Logistic Regression, and Support Vector Machine (SVM). SVM achieved the highest held-out test accuracy at **95.83%**, while Naive Bayes achieved the strongest five-fold training cross-validation macro F1 at **97.01%** and was selected for the integrated prototype.

Beyond classification, NewsBot extracted 40,803 named entity mentions, analyzed sentiment patterns across news categories, examined grammatical and syntactic differences, and demonstrated an integrated processing workflow on nine sample articles.

My main finding is that **classification provides organization, while linguistic and contextual analysis provides the information needed to interpret and act on that organization**.

The results demonstrate a functional academic prototype with promising performance on the BBC dataset. They do not establish production readiness or reliable performance across current news sources.

2. Business Problem and Project Objectives

2.1 Business Problem

News organizations, media analysts, and businesses process large volumes of articles from different sources. Manual review requires time, and traditional keyword filtering may miss useful context.

For example, two articles can both belong to the Business category while discussing very different subjects. One may describe company expansion, while another focuses on financial losses or economic uncertainty.

A classification system can group both articles correctly, but additional analysis is needed to understand their content.

NewsBot addresses this problem by combining several NLP techniques within one processing workflow.

2.2 Target Users and Business Value

The prototype is designed around three potential use cases:

Target User	Potential Value
News editors and content managers	Automatically organize articles into categories
Media monitoring analysts	Identify frequently mentioned entities and sentiment patterns
Business intelligence teams	Explore organizations, topics, and contextual signals across news coverage

These are potential applications rather than measured business outcomes. The project does not quantify labor savings, financial returns, or operational efficiency improvements.

2.3 Project Objectives

The implementation focuses on four objectives:

1. Build a reproducible NLP pipeline that integrates techniques from Modules 1 through 8.
2. Compare multiple classification configurations using appropriate evaluation metrics.
3. Extract additional information through linguistic, sentiment, emotion, and entity analysis.
4. Translate technical findings into practical insights while documenting limitations.

My design priority was not to maximize model complexity. It was to build a system in which each component contributes a clear analytical purpose.

3. Dataset Acquisition and Preparation

3.1 Dataset Selection

I used the BBC News Classification Dataset available through Kaggle.

Source: https://www.kaggle.com/c/learn-ai-bbc/data

The dataset contains full English-language news articles with predefined category labels. It is appropriate for supervised multi-class classification and sufficiently small to process using Google Colab's free resources.

3.2 Dataset Statistics

Dataset Metric	Recorded Result
Original articles	1,490
Records removed for missing values	0
Exact duplicates removed	50
Final unique articles	1,440
News categories	5
Training articles	1,152
Testing articles	288

Training/test ratio	80% / 20%

The cleaned dataset contains the following category distribution:

Category	**Articles**	**Share**
Sport	342	23.75%
Business	335	23.26%
Politics	266	18.47%
Entertainment	263	18.26%
Technology	234	16.25%
Total	**1,440**	**100%**

The categories are not perfectly balanced, but each contains enough articles for training and evaluation. Stratified sampling was used to preserve category proportions across the train/test split.

3.3 Data Cleaning and Preprocessing

The preprocessing workflow includes:

- Validating text and category columns.
- Removing missing records and exact duplicate content.
- Normalizing category labels.
- Cleaning and normalizing text.
- Tokenization and stopword removal.
- Lemmatization.
- Preparing features for classification and linguistic analysis.

A key design consideration was preventing information leakage. The classification pipeline fits TF-IDF and applicable numerical transformations within the training process. Cross-validation fits these transformations separately within each training fold.

This prevents vocabulary and scaling information from validation folds from being used during model fitting.

3.4 Dataset Limitations

The BBC dataset does not contain original publication dates or separate, authentic article headlines. The notebook generates title proxies for certain analyses, but these are not actual published headlines.

As a result, the project cannot reliably evaluate temporal trends or headline-specific behavior.

The dataset also represents a limited news collection rather than the full diversity of current journalism.

These limitations affect how the findings should be interpreted, especially when considering future real-world deployment.

4. System Architecture and NLP Integration

NewsBot follows a modular processing architecture.

Input: Raw news article

Processing:

1. Text cleaning and normalization.
2. TF-IDF feature generation.
3. Linguistic feature analysis.
4. Category classification.
5. Sentiment and emotion analysis.
6. Named entity extraction.
7. Dependency and syntax analysis.

Output: Structured article insights, including the predicted category, model-estimated probabilities, sentiment, named entities, syntactic information, and interpretable observations.

The classification component uses TF-IDF features, with additional numerical features incorporated in the Logistic Regression and SVM configurations.

Other modules operate as complementary analysis components.

This distinction matters. Not every analytical feature is used directly to train the final classifier. Instead, NewsBot combines predictive modeling with separate methods that help describe and interpret article content.

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5. Implementation of the Eight NLP Modules

5.1 Module 1: Real-World NLP Application Context

The project establishes a news intelligence use case focused on article organization, information discovery, and media monitoring.

The business context defines why the system is useful and how different NLP outputs could support user decisions.

5.2 Module 2: Text Preprocessing

The preprocessing module converts raw article text into a more consistent representation.

Cleaning, tokenization, stopword removal, and lemmatization reduce unnecessary variation in text features.

Removing exact duplicates also helps reduce the risk of identical articles appearing in both training and testing datasets.

The resulting 1,440-article dataset provides the foundation for the remaining modules.

5.3 Module 3: TF-IDF Feature Extraction

TF-IDF represents text by considering how frequently a term appears within an article and how distinctive it is across the document collection.

The implementation examines unigrams, bigrams, and category-specific terms.

The saved analysis identifies these representative terms:

Category	Representative Terms
Business	firm, company, market, year, share
Entertainment	film, star, award, best, actor
Politics	labour, election, party, government, blair
Sport	game, win, play, player, match
Technology	use, people, user, technology, computer

These results demonstrate that different news categories have recognizable vocabulary patterns.

For example, Business articles frequently contain terms related to companies and markets, while Sport articles emphasize games, players, and competitive outcomes.

However, TF-IDF identifies statistical term importance rather than the complete meaning of a sentence.

This is why the project also includes syntax, sentiment, and entity analysis.

5.4 Module 4: Part-of-Speech Analysis

Part-of-speech tagging identifies grammatical functions such as nouns, verbs, adjectives, and numerical expressions.

The saved analysis reports:

Linguistic Pattern	Highest Category
Noun-related pattern	Entertainment
Verb-related pattern	Politics
Number-related pattern	Business

These findings suggest that grammatical patterns differ across the BBC news categories.

Business articles showing stronger numerical patterns is consistent with reporting that frequently discusses financial values, market figures, and quantitative information.

The result is descriptive. It does not establish that all business journalism follows the same writing style.

5.5 Module 5: Syntax Parsing and Semantic Analysis

The syntax module uses dependency parsing to examine grammatical relationships within sentences.

The implementation analyzes sentence structure, noun phrases, action patterns, and dependency-based features.

The saved results identify:

- **Politics:** Highest sentence-length pattern.
- **Sport:** Highest dependency-distance pattern.

These findings show that categories can differ in sentence construction, not only in vocabulary.

This is important because two articles may share similar words but use different grammatical structures to communicate information.

Dependency parsing adds structural context that basic keyword analysis cannot provide.

5.6 Module 6: Sentiment and Emotion Analysis

NewsBot uses VADER for sentiment analysis and NRC-based lexical emotion associations.

VADER estimates positive, negative, and neutral sentiment and provides a compound polarity score.

The recorded average compound scores are:

Category	Average VADER Compound Score
Entertainment	0.590
Sport	0.549
Technology	0.533
Business	0.238
Politics	0.054

Entertainment has the highest average compound score, while Politics has the lowest.

The saved category-level distribution also reports that 46.62% of Politics articles were classified as negative, compared with 19.01% of Entertainment articles.

This is one of the most interesting findings in the project, but it requires careful interpretation.

A lower sentiment score does not automatically mean biased or misleading reporting.

Political reporting may discuss conflicts, disagreements, public concerns, or negative events. These subjects can influence lexicon-based sentiment scores without demonstrating bias.

For emotion analysis, the notebook evaluates 150 sampled articles, with 30 from each category, using eight NRC emotion associations:

Anger, Anticipation, Disgust, Fear, Joy, Sadness, Surprise, and Trust.

All 150 sampled articles contained at least one matched emotion association in the saved experiment.

These results represent word-level emotion associations. They do not prove the emotional intent of an author or the emotional reaction of a reader.

The distinction is important when translating NLP output into business intelligence.

5.7 Module 7: Multi-Class Text Classification

Three classification configurations were evaluated using a stratified 80/20 train/test split and five-fold cross-validation.

The primary model-selection metric was training cross-validation macro F1.

Recorded model comparison:

Model	Test Accuracy	Test Macro F1	CV Macro F1
Naive Bayes	94.44%	94.03%	97.01%

Logistic Regression	92.71%	92.36%	90.99%
SVM	**95.83%**	**95.57%**	95.83%

SVM achieved the highest accuracy and macro F1 on the held-out test set.

However, Naive Bayes achieved the strongest five-fold cross-validation macro F1 on the training data.

I selected Naive Bayes for the integrated NewsBot prototype because model selection was based on training cross-validation results, not the held-out test results.

This avoids using the test set to choose the model after observing its performance.

An important limitation is that the configurations do not use identical feature sets. Therefore, the results compare complete model configurations rather than isolating the effect of the classification algorithms alone.

The results show that traditional machine learning methods can perform strongly on a structured, five-category news dataset without requiring computationally expensive deep learning models.

5.8 Module 8: Named Entity Recognition

NewsBot uses spaCy to extract named entities such as people, organizations, locations, dates, and monetary values.

The saved analysis reports:

NER Metric	Result
Total entity mentions	40,803
Unique entity strings	11,995
Entity types	18
Average entity mentions per article	28.34
Co-occurrence network nodes	30
Co-occurrence network edges	222

The system also analyzes entity frequencies and category-specific patterns.

Entity co-occurrence is used to identify entities mentioned within the same article.

This can help analysts discover which organizations, people, or locations frequently appear together.

However, **co-occurrence does not prove a real-world relationship**.

Two organizations appearing in the same article may be connected by a business transaction, a disagreement, or simply a shared news topic. The network only establishes shared article context.

Manual NER Error Analysis

The repository includes a 30-prediction manual review dataset.

The recorded examples demonstrate several important NER limitations:

- *Tony Blair* was correctly identified as PERSON.
- *William Hill* was incorrectly labeled PERSON instead of ORG.

- *Lisa Raymond 7-5 6-3* incorrectly included a tennis score within a person's name.
- *Ulrich* was incompletely extracted and incorrectly classified as ORG instead of PERSON.

These examples show that NER errors can involve both incorrect labels and incorrect entity boundaries.

Manual review is therefore valuable because a large number of extracted entities does not automatically indicate high extraction quality.

The 30-prediction review also cannot establish full NER recall or F1 because it does not contain exhaustive ground-truth annotations for complete articles.

6. Integrated System Testing

The completed NewsBot prototype accepts an article and produces a structured output combining classification and NLP analysis.

The saved demonstration includes nine examples:

- Seven synthetic articles.
- Two external-source articles.

All nine examples executed successfully.

The two external articles were sourced from NASA and the Federal Reserve.

External Article	Expected Category	Predicted Category
NASA technology collaboration announcement	Technology	Technology
Federal Reserve monetary policy announcement	Business	Business

Both external predictions matched their manually assigned categories.

However, two correctly classified external articles are not sufficient to establish reliable generalization accuracy.

The test results also show that category probability estimates vary across examples, including lower-confidence predictions for ambiguous content.

These probabilities are model estimates and have not been demonstrated to be calibrated.

The integrated tests confirm that the components can produce combined outputs on the saved examples. They do not demonstrate production reliability across different publishers, time periods, or article styles.

7. Key Findings and Business Insights

Finding 1: Strong classification does not equal complete news understanding

The classification results demonstrate that TF-IDF-based machine learning can identify BBC news categories with high accuracy.

However, a category label does not explain the entities, events, tone, or grammatical relationships within an article.

The additional NLP modules provide complementary information that makes the classification output more interpretable.

Finding 2: News categories have measurable linguistic differences

The TF-IDF, POS, and syntax analyses identify differences in vocabulary and writing patterns across the five categories.

These patterns help explain why classification works and show how linguistic analysis can reveal characteristics that are not visible from category labels alone.

Finding 3: Sentiment is a signal, not a judgment

The Politics category has the lowest average VADER sentiment score in the saved analysis.

This could help a media analyst identify content requiring further review, but it should not be interpreted as evidence of political bias.

Sentiment scores describe characteristics of the analyzed text under a particular method. They do not independently establish truthfulness, fairness, or intent.

Finding 4: Entity extraction creates opportunities for information discovery

NewsBot extracted more than 40,000 entity mentions and constructed a co-occurrence network.

This demonstrates the potential for entity-based article discovery and monitoring.

At the same time, the manual review highlights extraction errors that must be addressed before these outputs can support high-stakes decisions.

Finding 5: Model selection should follow a defined evaluation process

Although SVM achieved the highest held-out test accuracy, Naive Bayes was selected based on training cross-validation macro F1.

This illustrates an important machine learning principle: evaluation methodology matters as much as the reported accuracy.

The goal is not simply to choose whichever model produces the largest number after testing.

8. Technical Challenges and Limitations

8.1 Integrating Different NLP Techniques

One challenge was combining methods that produce different types of information.

TF-IDF generates numerical text features, POS tagging produces grammatical labels, dependency parsing identifies structural relationships, sentiment analysis estimates polarity, and NER extracts entity spans.

The implementation addresses this by organizing the methods into complementary processing components rather than treating every output as a classification feature.

8.2 Data Leakage and Evaluation

Classification pipelines must prevent information from the evaluation data from influencing model training.

The project uses stratified splitting and fits preprocessing transformations within the training and cross-validation processes.

This supports a more reliable evaluation than fitting TF-IDF to the complete dataset before splitting.

8.3 Limited External Validation

The held-out BBC test results are encouraging, but the external demonstration contains only two independently sourced articles.

More external evaluation is needed before drawing conclusions about performance on current news.

8.4 Interpretation of Linguistic Results

Sentiment scores, emotion associations, and entity co-occurrence can be useful analytical signals, but they have limitations.

Treating these outputs as verified facts could produce misleading conclusions.

The system therefore requires human interpretation when results are used for business decisions.

8.5 Dataset and Resource Constraints

The BBC dataset lacks publication dates and authentic separate headlines, limiting temporal and headline analysis.

The project was also designed for Google Colab free-tier resources, so computational efficiency and manageable dataset size were important considerations.

9. Individual Contributions and Development Approach

I completed this project independently, including the dataset preparation, NLP implementation, model comparison, analysis, evaluation, and GitHub documentation.

My work covered the complete development process:

1. Reviewing the assignment requirements and defining the system scope.
2. Selecting and preparing